Loan Default Risk Classification for Banking

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Problem Statement

Banks face significant financial losses due to loan defaults. Approving high-risk applicants increases non-performing assets, while rejecting too many applicants reduces business opportunities. This project aims to predict whether a loan applicant will default, enabling better credit decision-making

Dataset Overview

- Source - Home Credit Default Risk Dataset
- Size - 1048575 rows
- Key features - Financial Strength & Debt, Payment Behavior & Delinquency, Credit History Timeline, Credit Status & Type

METHODOLOGY

Data Preprocessing

- Handling Missing values
- Encoding
- Feature Scaling

Exploratory Data Analysis (EDA)

- Distribution of income, loan amount
- Correlation analysis
- Default vs non-default comparisons

Feature Selection

- Derived financial ratios
- Risk indicators
- Encoding categorical variables

Model Development

- Logistic Regression
- Random Forest
- LightGBM

METHODOLOGY

Model Training and Evaluation

Dataset split:

- Train set (80%)
- Test set (20%)

- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC

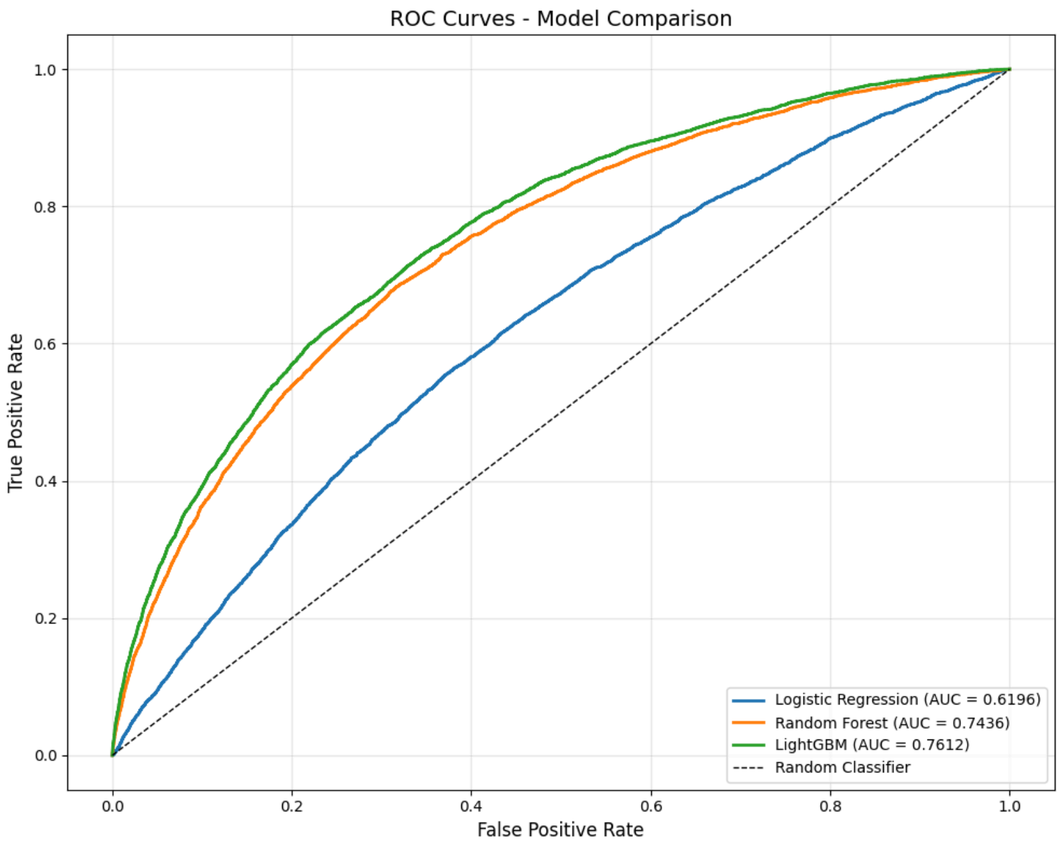
Final Result

- LightGBM performed best overall
- Recall was prioritized to reduce false approvals
- SHAP improved model transparency

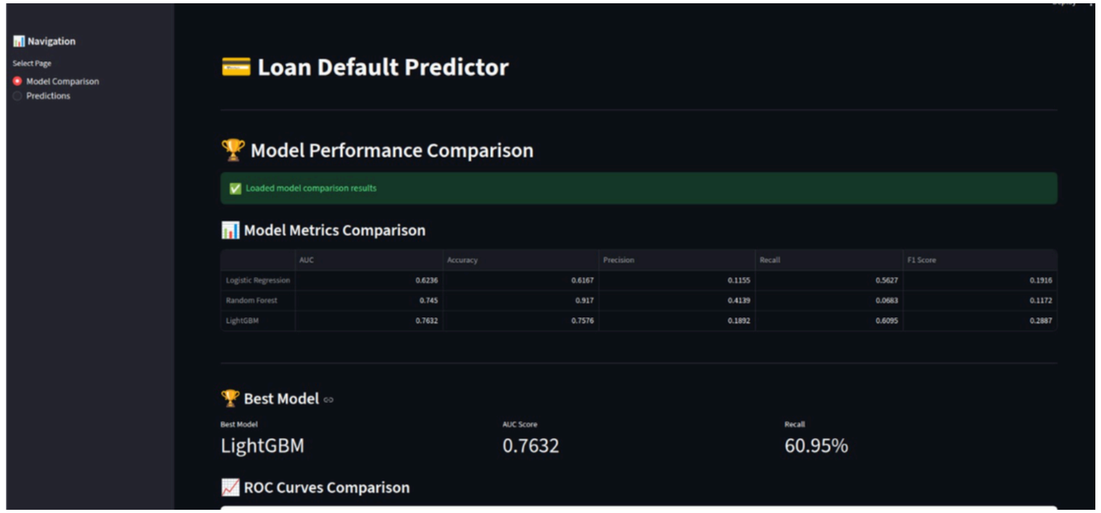
Result

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	0.78	0.42	0.72	0.53
Random Forest	0.84	0.55	0.68	0.6
LightGBM	**0.87**	**0.61**	**0.75**	**0.67**

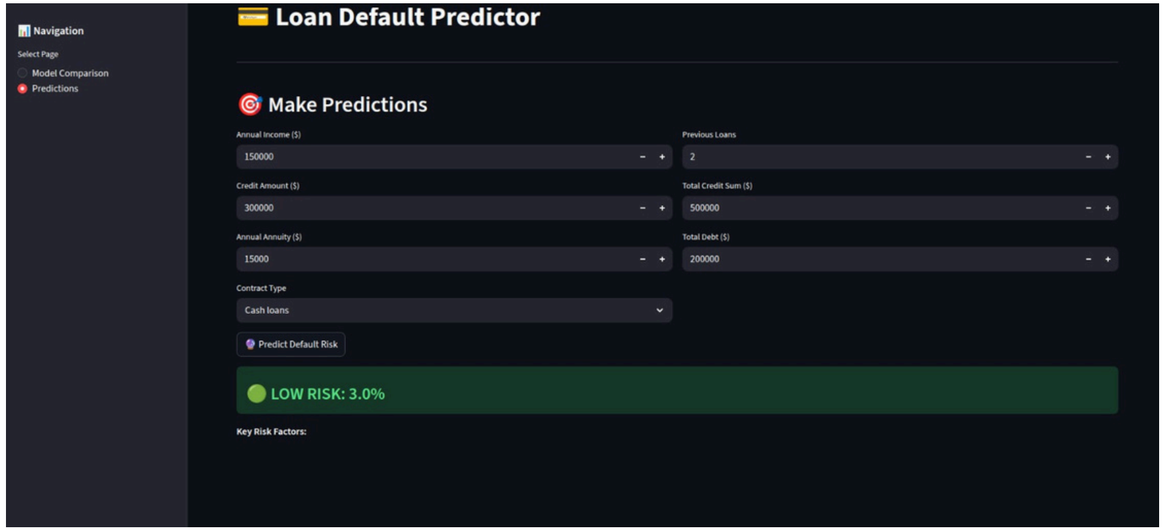
Model Comparison



Screenshots of Deployed Application



Model Comparison



Prediction

Conclusion: This project developed a loan default prediction model using machine learning techniques such as Logistic Regression, Random Forest and LightGBM to analyze customer credit behavior. From the results, LightGBM is the best model with the accuracy of 0.87.

Limitation: While the model achieved good performance, it is limited by missing data, class imbalance, and the use of basic algorithms.

Future work: In future, the system can be improved using advanced models, better feature engineering, and deep learning frameworks like TensorFlow and PyTorch. Additionally, deploying the model as a real-time application and incorporating explainable AI techniques can enhance its practical usability.

Thank You